A probabilistic merit order for electricity price forecasting

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University of Duisburg-Essen
University of Glasgow

Academy of Economic Studies (ASE), Bucharest, 03-05.06.2026











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Overview

Introduction & Motivation

1. Classical merit order

2. Data-driven merit order

3. Probabilistic merit order

Backup slides

Paul Ghelasi

Current Position

- Research Associate at the **University of Duisburg-Essen**
  - **House of Energy, Climate and Finance**
  - **Chair of Data Science in Energy and Environment**

Academic Career

- **B.Sc. Business Administration (Finance)**, University of Mannheim
- **M.Sc. Energy and Finance**, University of Duisburg-Essen
- **PhD** (ongoing), University of Duisburg-Essen
- **Visiting Researcher**, University of Glasgow, UK
- **Visiting Lecturer**, National University of Science and Technology Politehnica, Bucharest, Romania

Teaching

- Lecturer at UDE of tutorial on **"Econometrics of Electricity Markets"**
- Visiting lecturer at Politehnica on **"Data Science in Energy"**

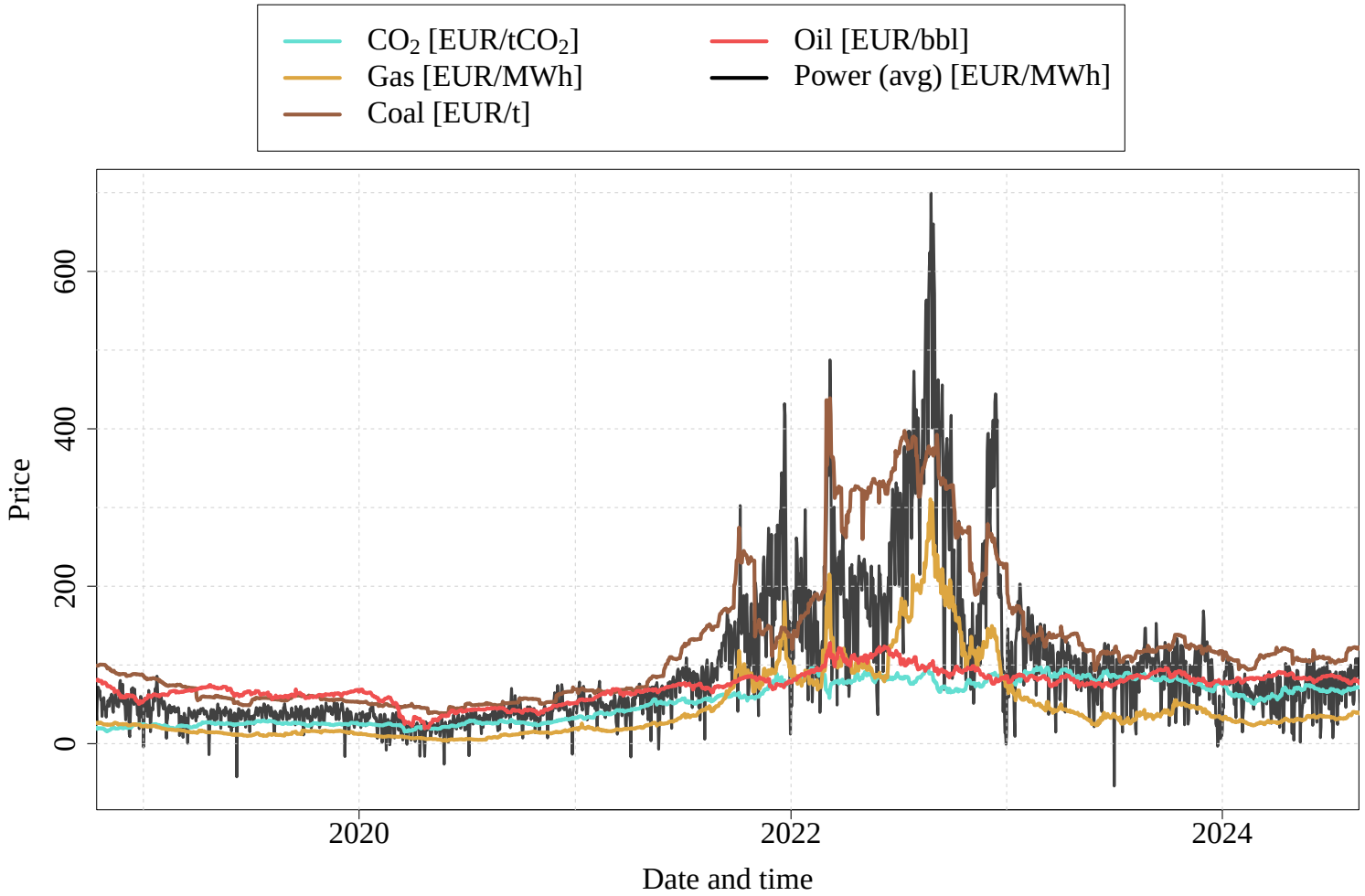
Research

- Focus areas: Electricity markets, price forecasting, energy trading
- Publications in: Energy Economics, International Journal of Forecasting, Renewable and Sustainable Energy Reviews

Professional Experience

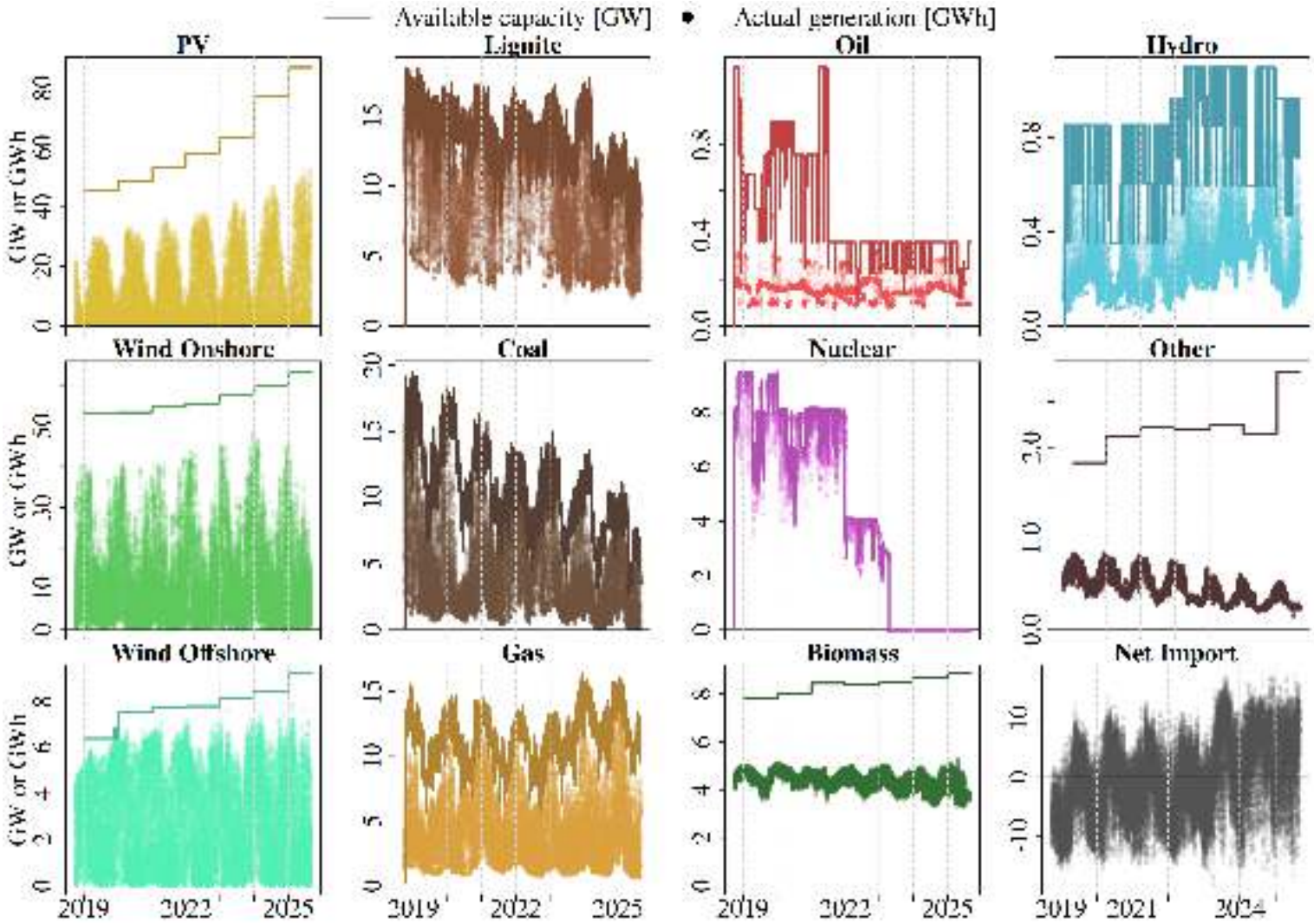
- **Credit Risk, Corporate Strategy, Trading strategy optimization**
- **@ RWE Supply & Trading**, **RWE Renewables**, industry projects

German day-ahead power, fuels and CO2 prices

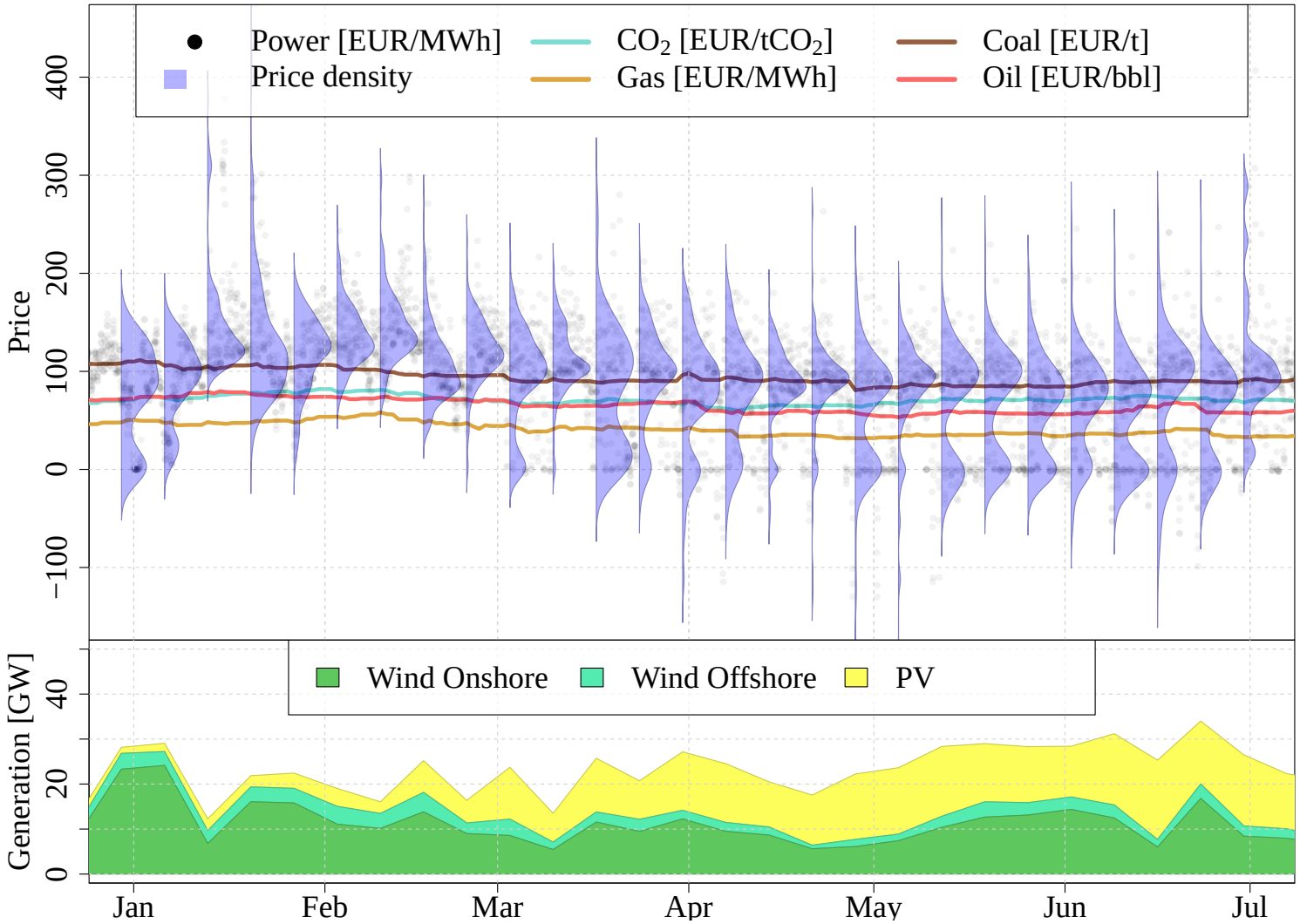


- Highly complex and volatile time series

German available capacities, generation and net imports

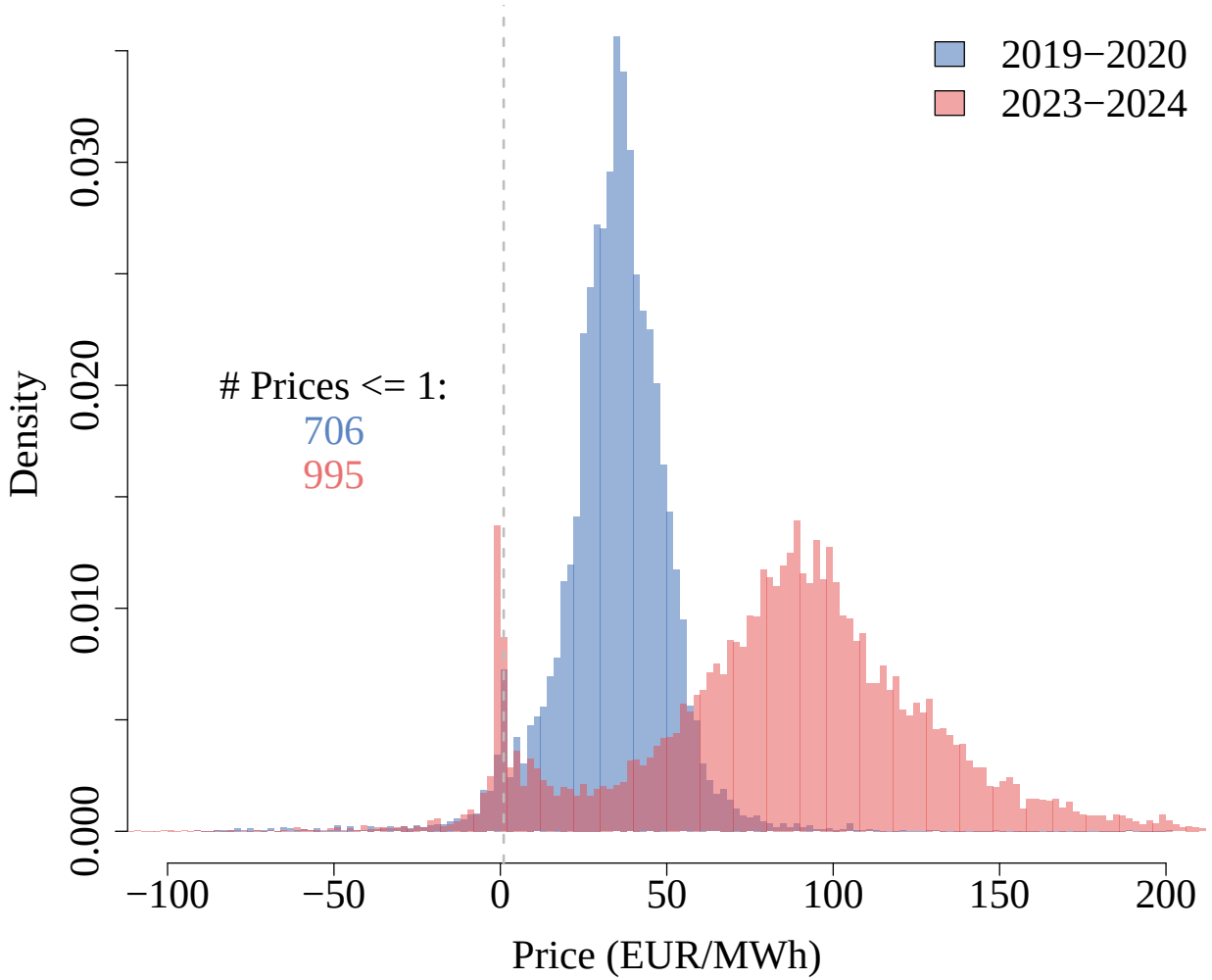


German day-ahead power price distributions



- Uuncertainty is dynamic, many sources of uncertainty

Histogram of German day-ahead power prices



- Bimodal gap widens => regime switch between high and low RES prices

Data-driven and fundamental models



Data-driven models
(Statistical, econometric, machine learning models)

Inputs:
Historical data

Outputs:
Power price

Data-driven and fundamental models



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(Statistical, econometric,
machine learning models)

Inputs:
Historical data

Outputs:
Power price

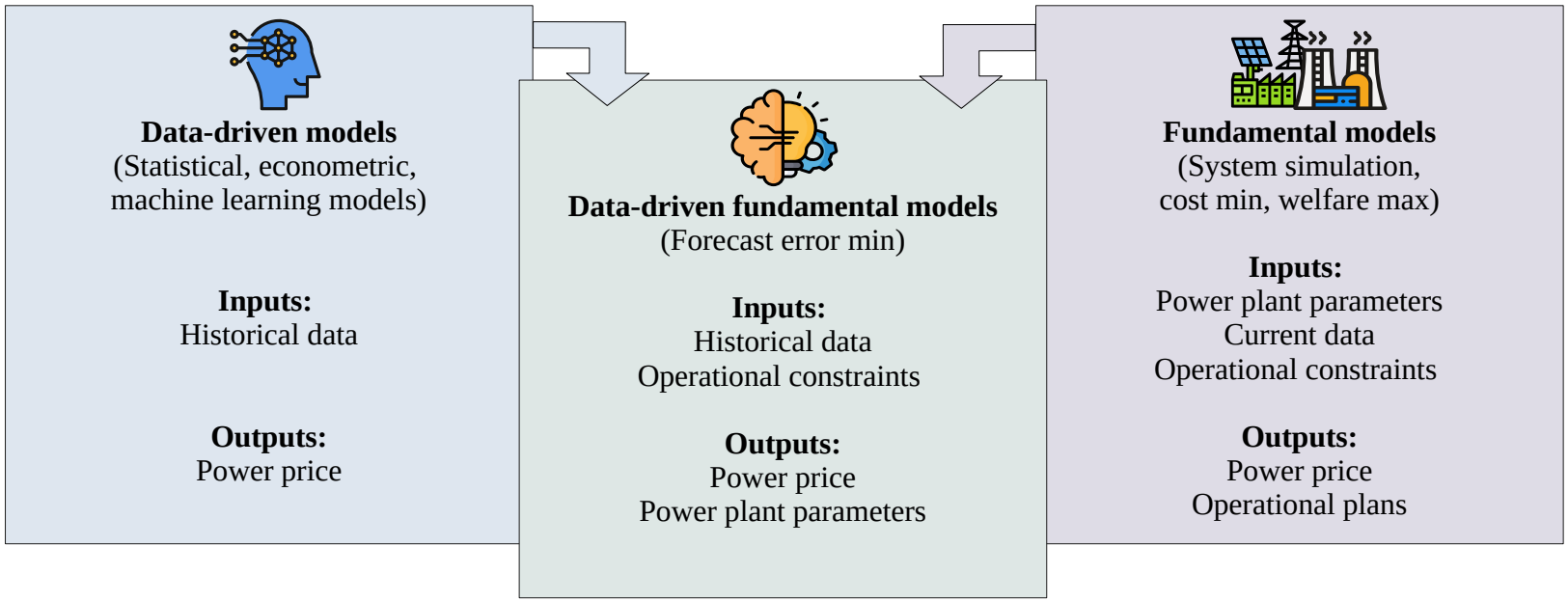


Fundamental models
(System simulation,
cost min, welfare max)

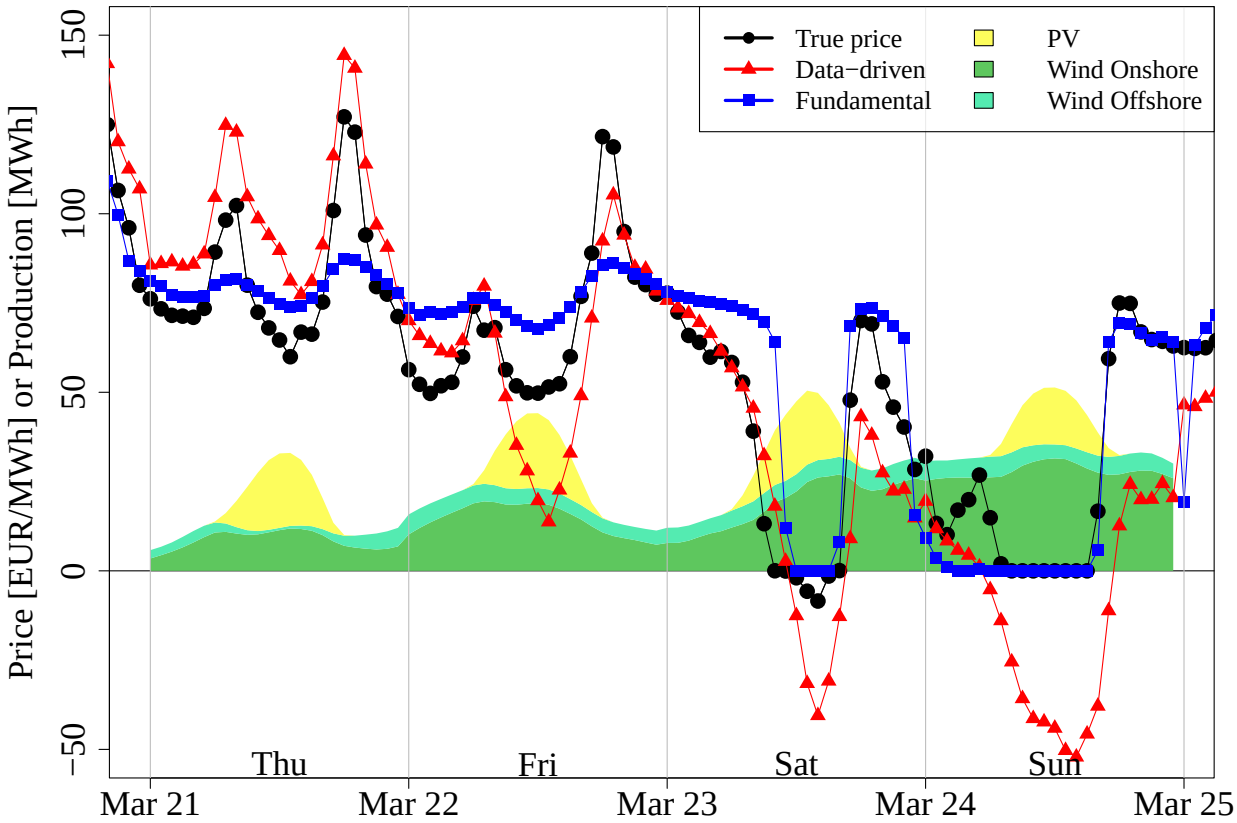
Inputs:
Power plant parameters
Current data
Operational constraints

Outputs:
Power price
Operational plans

Data-driven and fundamental models



Data-driven and fundamental models



- **Data-driven models**: capture volatility, but overshoot, no restrictions
- **Fundamental models**: capture price clusters, but too "flat"

Overview

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2. Data-driven merit order

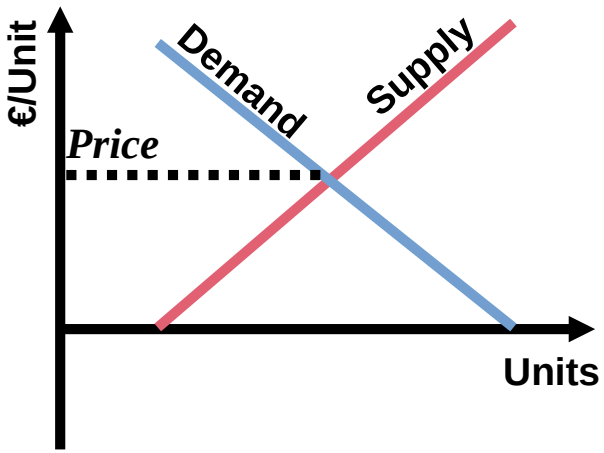
3. Probabilistic merit order

Backup slides

Simple fundamental model: the classical merit order

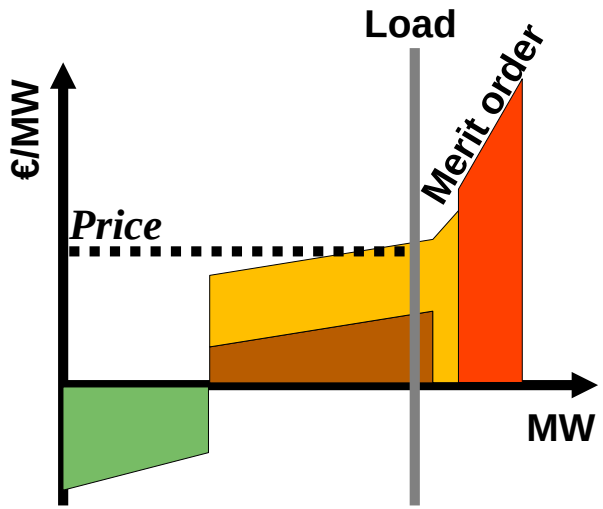
Economics

Price is at the intersection of supply and demand



Energy economics

Price is at the intersection of the merit order (supply stack) and load



Constructing the merit order

Constructing the merit order

Constructing the merit order

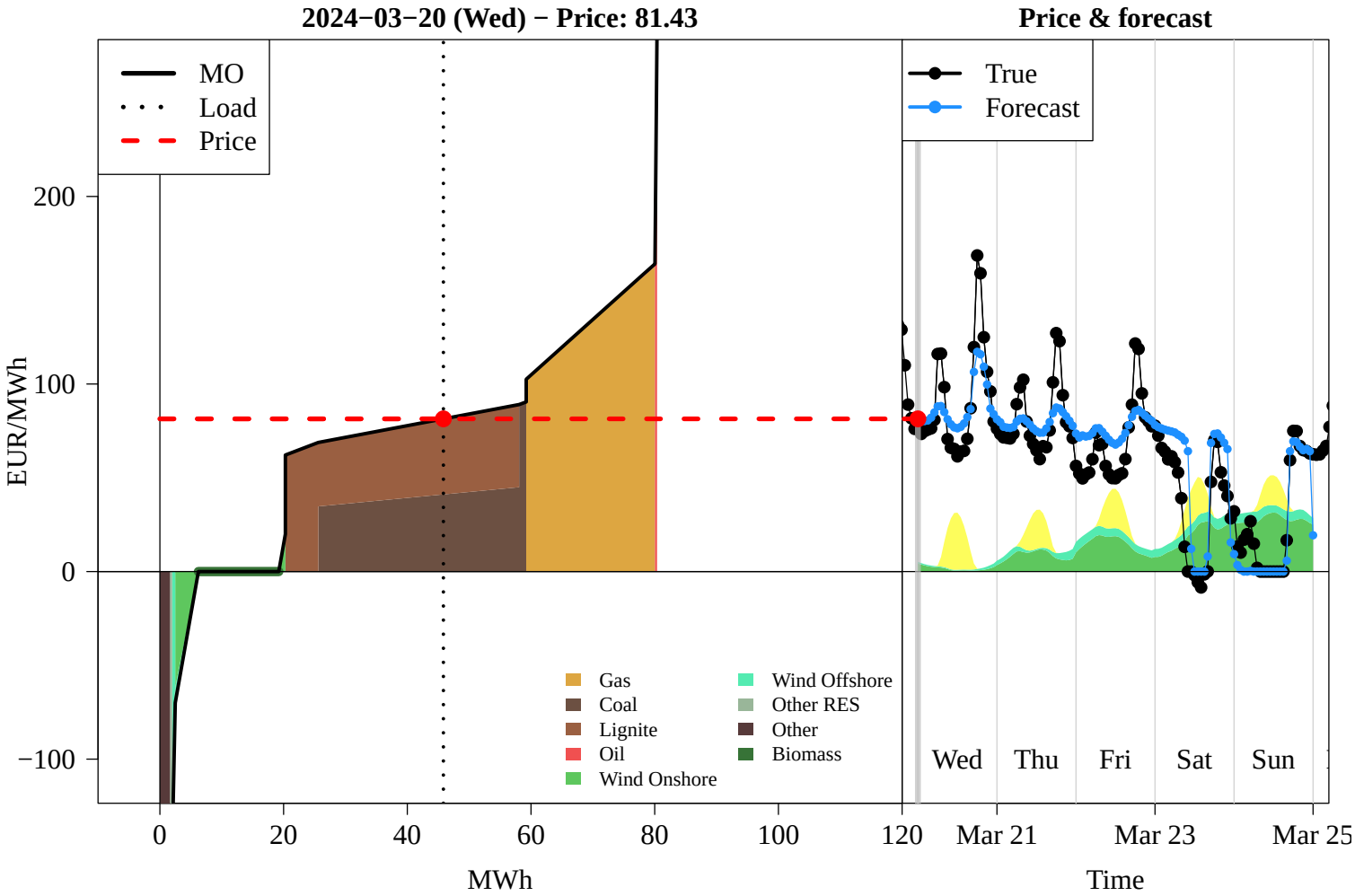
Constructing the merit order

Constructing the merit order

Constructing the merit order

Constructing the merit order

Modelling the price with the merit order

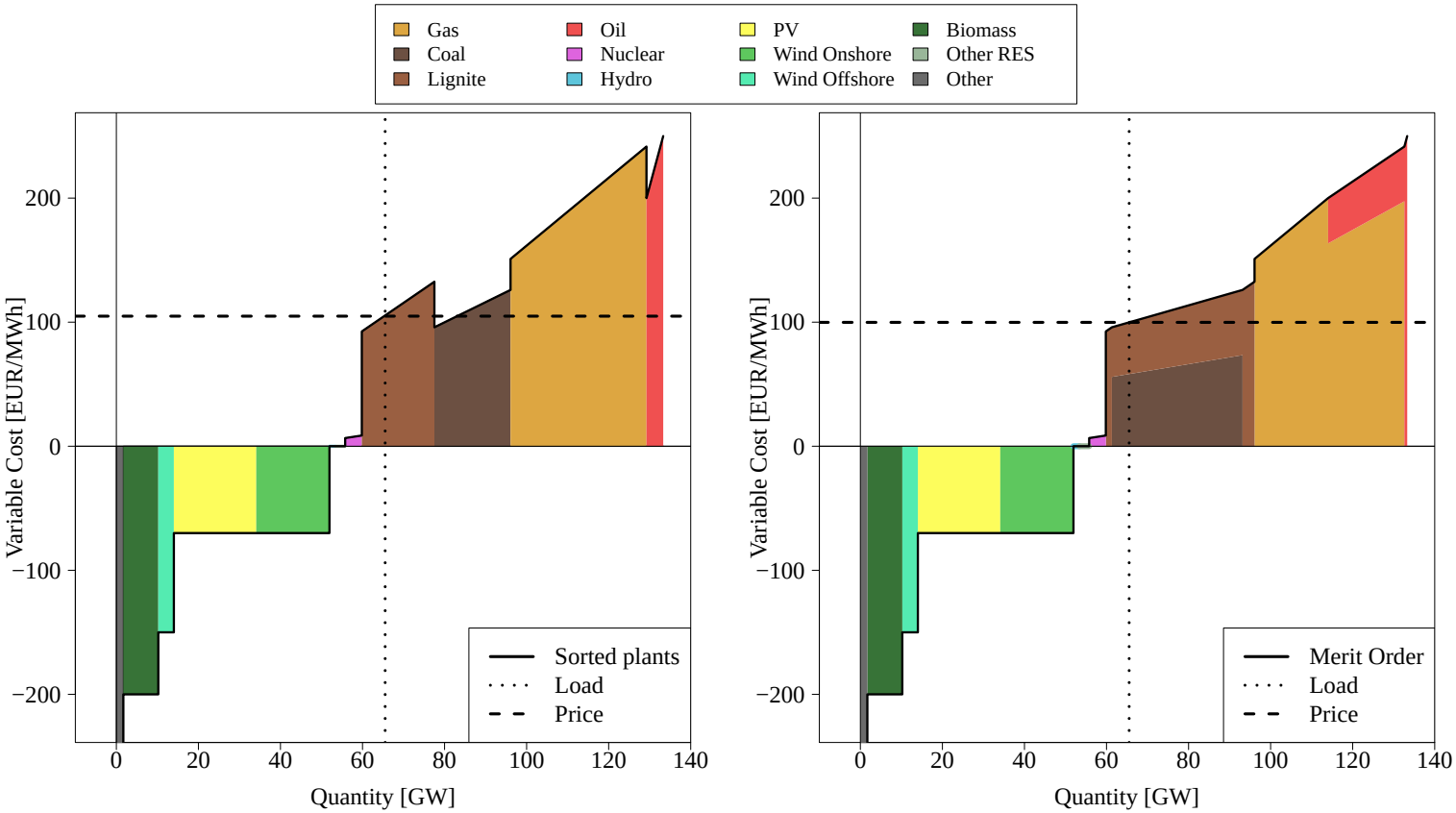


Expert estimates for efficiencies and CO2-intensity factors

	Lignite	Coal	Gas	Oil	Nuclear
Efficiency low $\left(\eta_{\text{pl}}^{\text{L}}\right)$	0.30	0.35	0.25	0.24	0.32
Efficiency high $\left(\eta_{\text{pl}}^{\text{U}}\right)$	0.43	0.46	0.40	0.44	0.42
$\mathbf{CO_2}$ **intensity** $\left(\varepsilon_{\text{pl}}\right)$	0.40	0.30	0.20	0.30	0.03

Table 1: Expert estimates for efficiencies and $\mathrm{CO_2}$-intensity factors for plants.[1]

Proper vs improper merit order



(a) Improper merit order

(b) Proper merit order

- To account for overlaps, the stacks must be inverted, summed up and un-inverted:

$$\mathrm{MO}^{-1}_{\mathrm{pl},t}(p) = \mathrm{Cap}_{\mathrm{pl},t}\,\frac{\max(\min(p,\mathrm{VarCost}^{\mathrm{U}}),\mathrm{VarCost}^{\mathrm{L}}) - \mathrm{VarCost}^{\mathrm{L}}_{\mathrm{pl},t}}{\mathrm{VarCost}^{\mathrm{U}}_{\mathrm{pl},t} - \mathrm{VarCost}^{\mathrm{L}}_{\mathrm{pl},t}} \qquad \mathrm{MO}_t(q) = \left(\sum\nolimits_{\mathrm{pl}\in\mathcal{T}} \mathrm{MO}^{-1}_{\mathrm{pl},t}\right)^{-1}(q)$$

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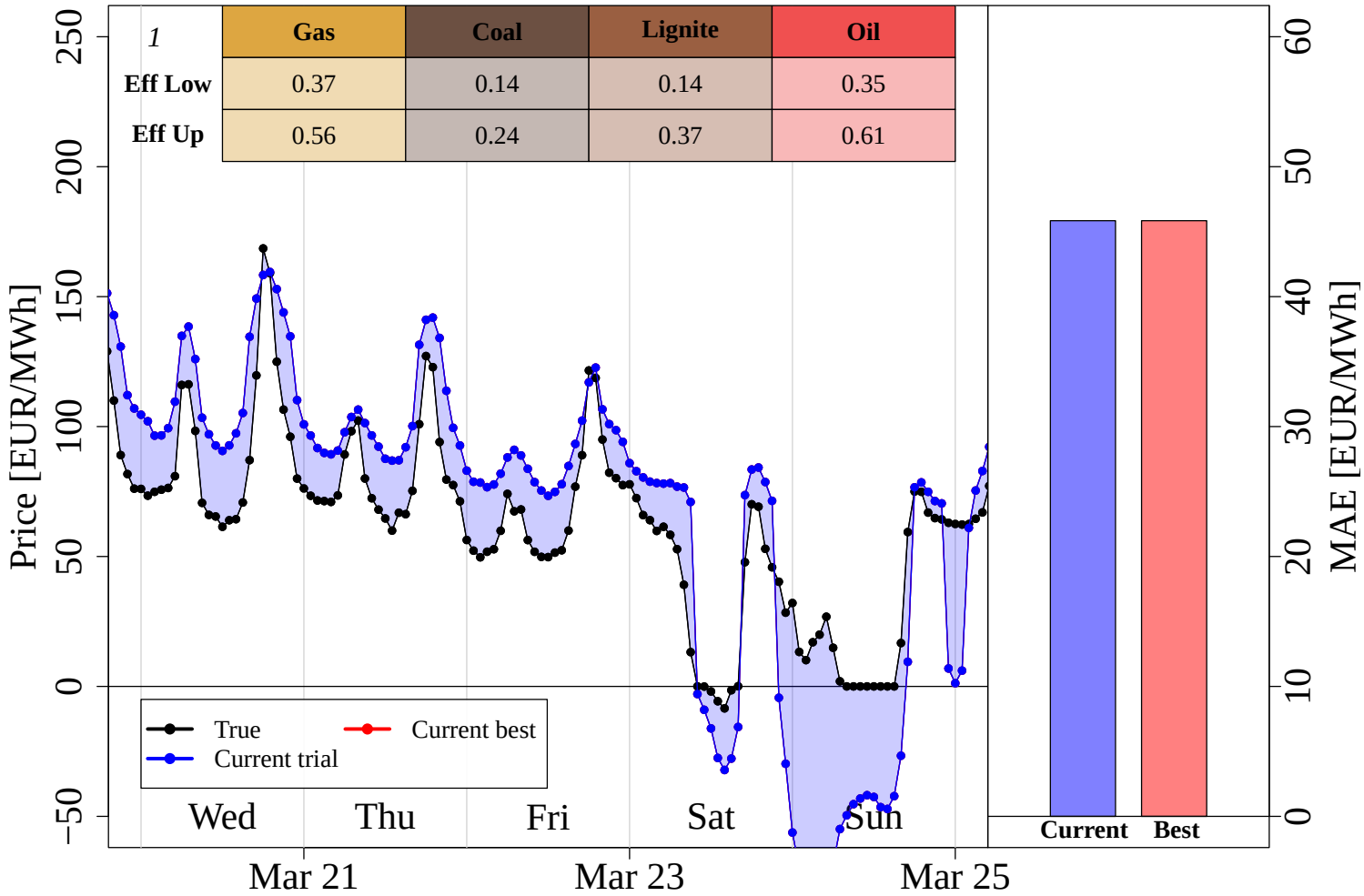
3. Probabilistic merit order

Backup slides

Estimating the efficiencies

Estimating the efficiencies

Estimating the efficiencies



Results

- Study design periods:

Training	Test
01/10/2022 - 01/10/2023	01/10/2023 - 01/10/2024

- No recalibration, parameters fixed for test period

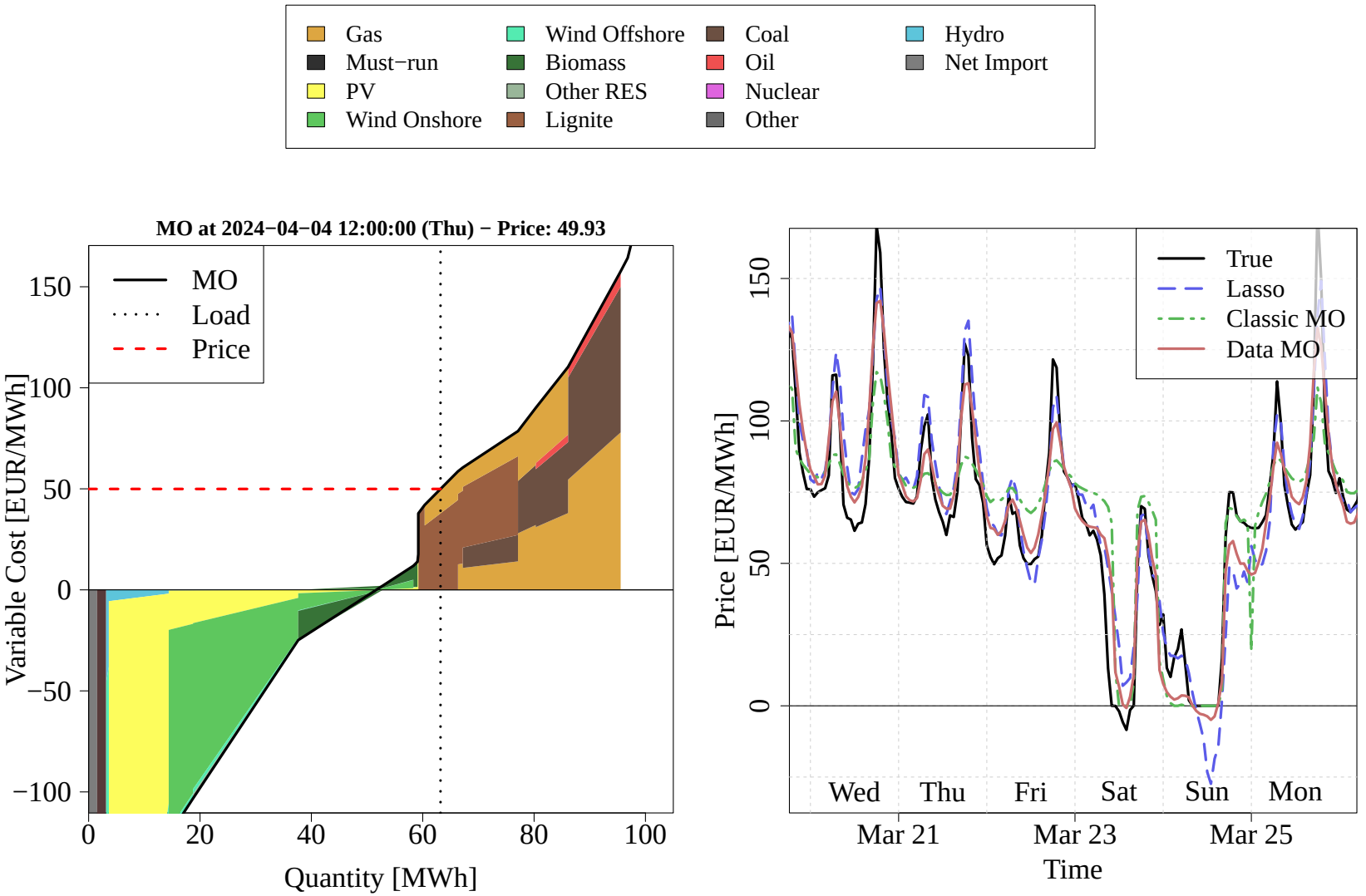
- Test errors MAE (EUR/MWh):

Model	Naive	Lasso	XGBoost	NNet	Classic MO	Data MO
Error	24.96	13.11	13.96	18.56	16.36	**11.83**
Skill	100%	53%	56%	74%	66%	**47%**

- Computation time statistics of a single run of the extended merit order model. One run calculates prices for 8760 hours and for 14 technology types.

	Min	LQ	Mean	Median	UQ	Max	Neval
Model run time (s)	0.41	0.58	0.74	0.71	0.88	1.45	100

Results



(c) Data-driven MO model

(d) Forecasts comparisons (2024)

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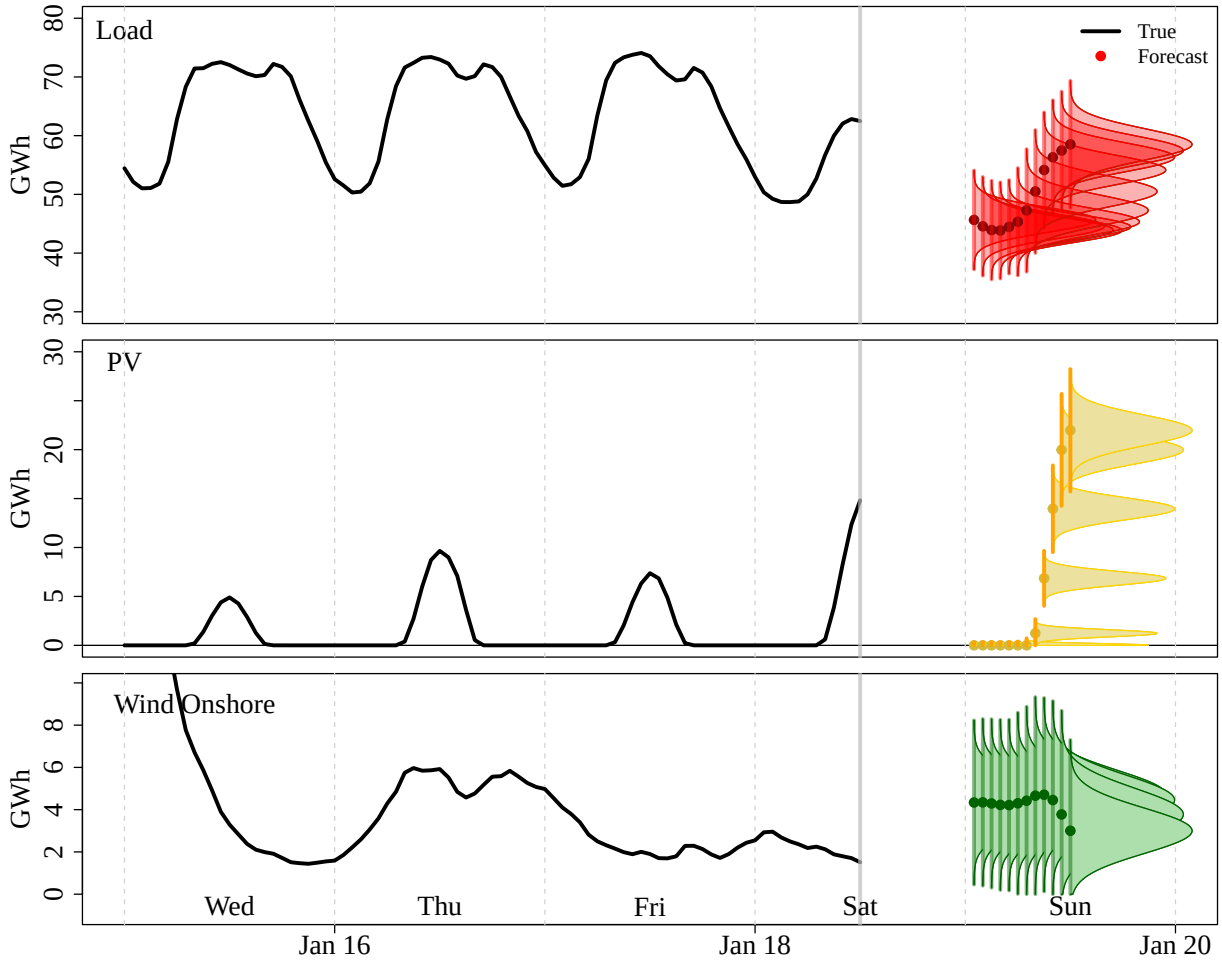
2. Data-driven merit order

3. Probabilistic merit order

Backup slides

Probabilistic merit order

- Fit a distributional model to each input of the merit order model, except for the parameters Θ.



Distributional forecasting of inputs (GAMLSS framework)

- Model each input's y_{t+1} distribution $\mathcal{D}$ given regressors $\mathbf{X}_t$

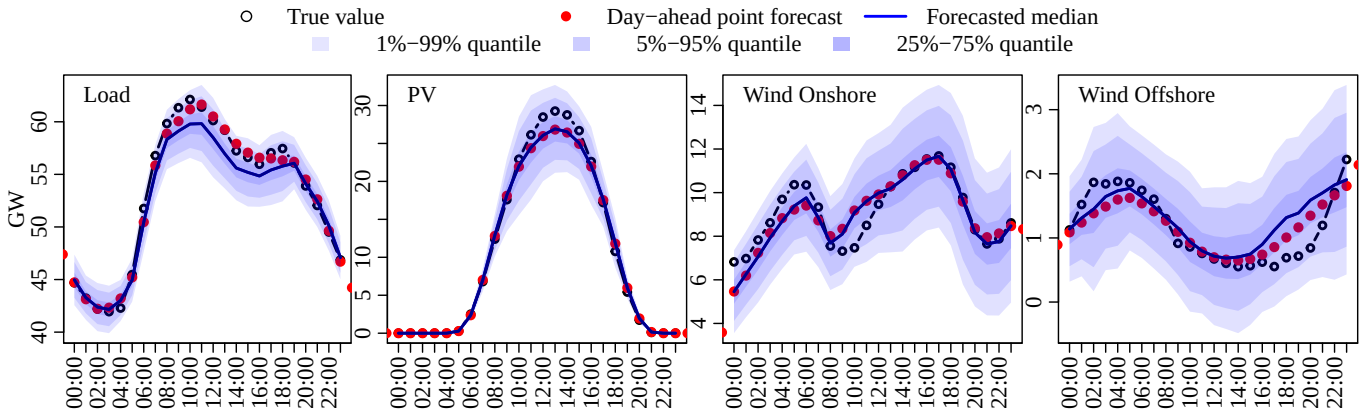
$$y_{t+1} \mid \mathbf{X}_t \sim \mathcal{D}\left(\theta_{t+1}^{(1)}, \ldots, \theta_{t+1}^{(n)}\right), \tag{1}$$

$$g_i\left(\theta_{t+1}^{(i)}\right) = \mathbf{X}_t \beta^{(i)}, \qquad i = 1, \ldots, n,$$

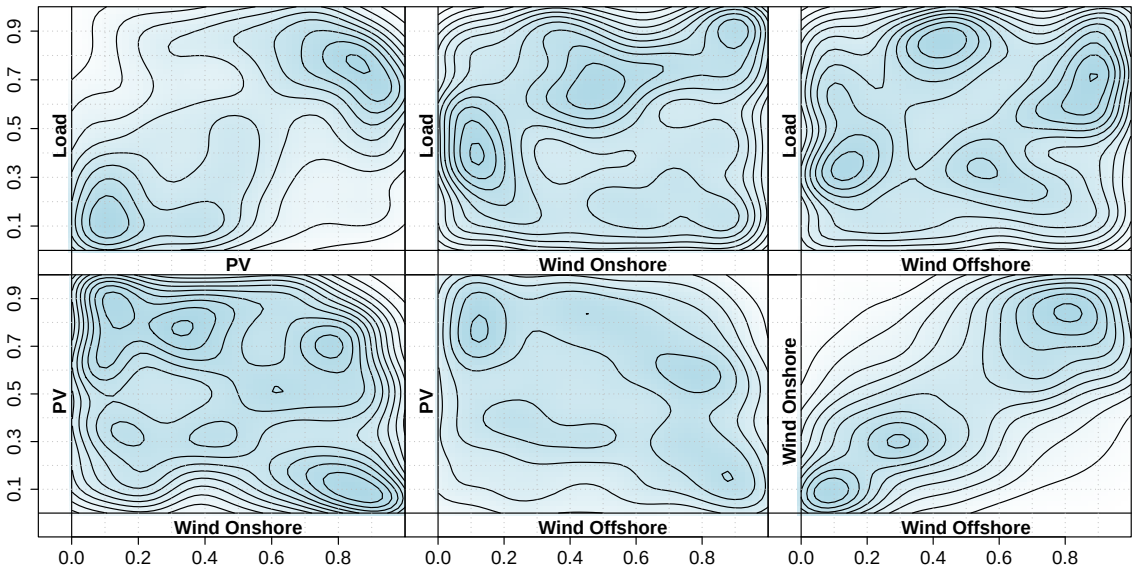
- e.g. $\mathcal{D} = \mathcal{N}$, $g(\cdot)$ is a link function

Input forecasting - results

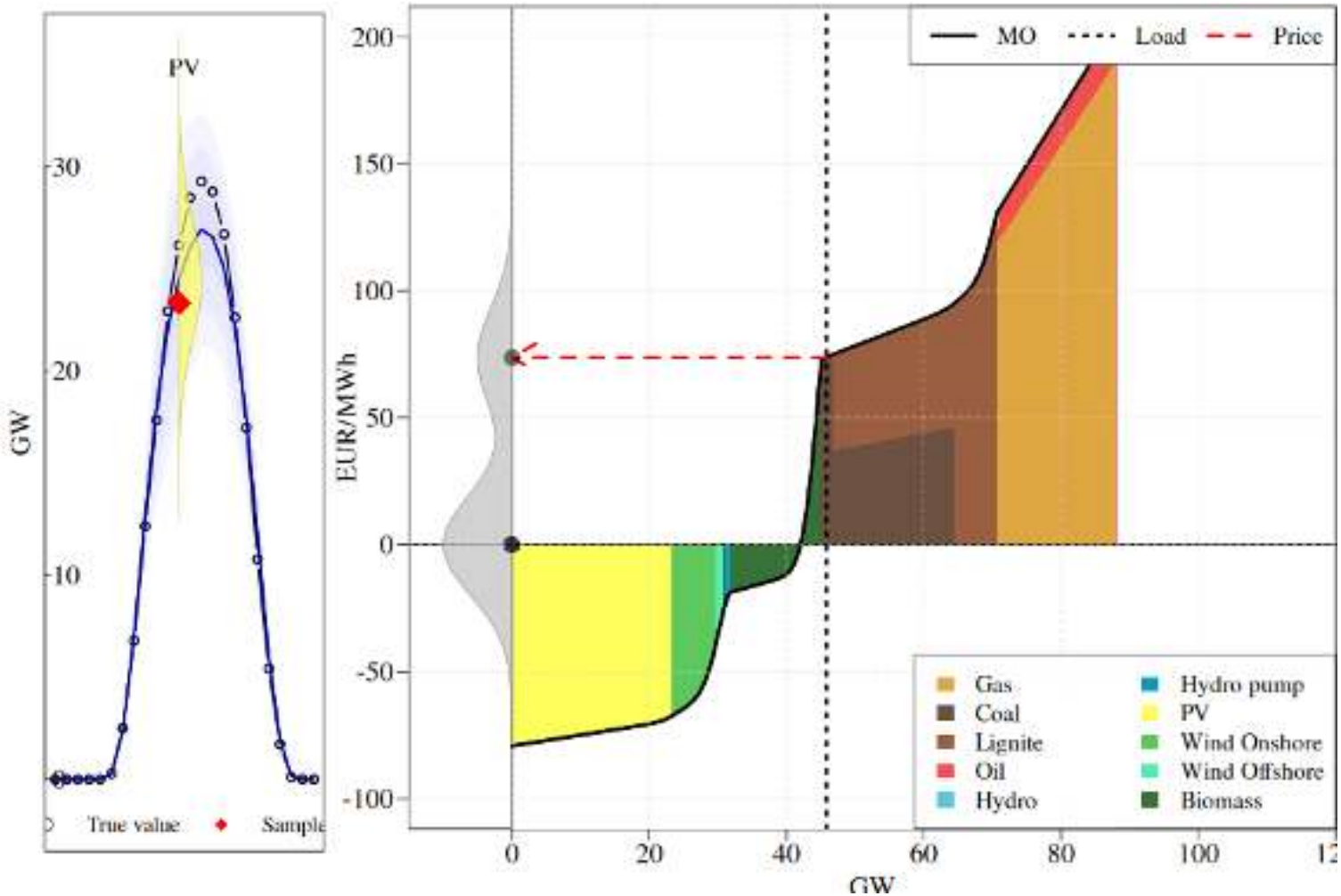
- Results: hourly distributional forecasts for all inputs



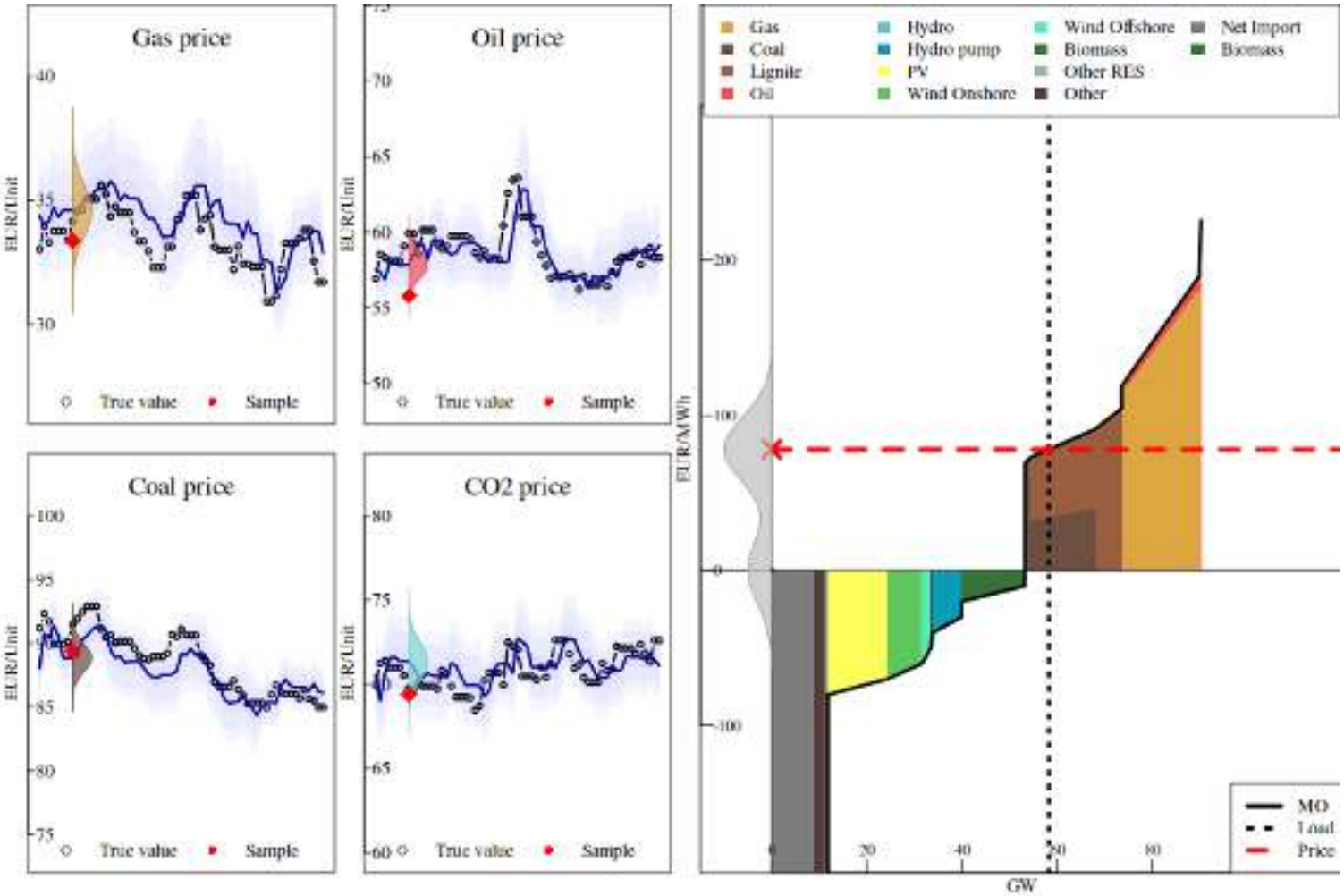
- Dependency modelling via empirical copula:



Sample& plug-in - RES



Sample& plug-in - Fuels



Estimating the parameters

- Parameters of the merit order model in:
  $\Theta^{\mathrm{MO}} = \{\text{Efficiencies}, \mathrm{CO2}\ \text{intensities}, \dots\}$
- Estimate the parameters Θ^{MO} by minimizing CRPS:

$$\hat{\Theta}^{\mathrm{MO}} = \underset{\Theta^{\mathrm{MO}}}{\arg\min} \left(\frac{1}{N} \sum_{t=1}^{N} \mathit{CRPS} \left(\underbrace{\mathrm{MO}_t(\mathrm{Load}_t; \Theta^{\mathrm{MO}})}_{\widehat{\mathrm{Price}}_t}, \mathrm{Price}_t \right) \right) \qquad (2)$$

$$\text{s.t.} \quad \Theta^{\mathrm{L}} \leq \Theta^{\mathrm{MO}} \leq \Theta^{\mathrm{U}}$$

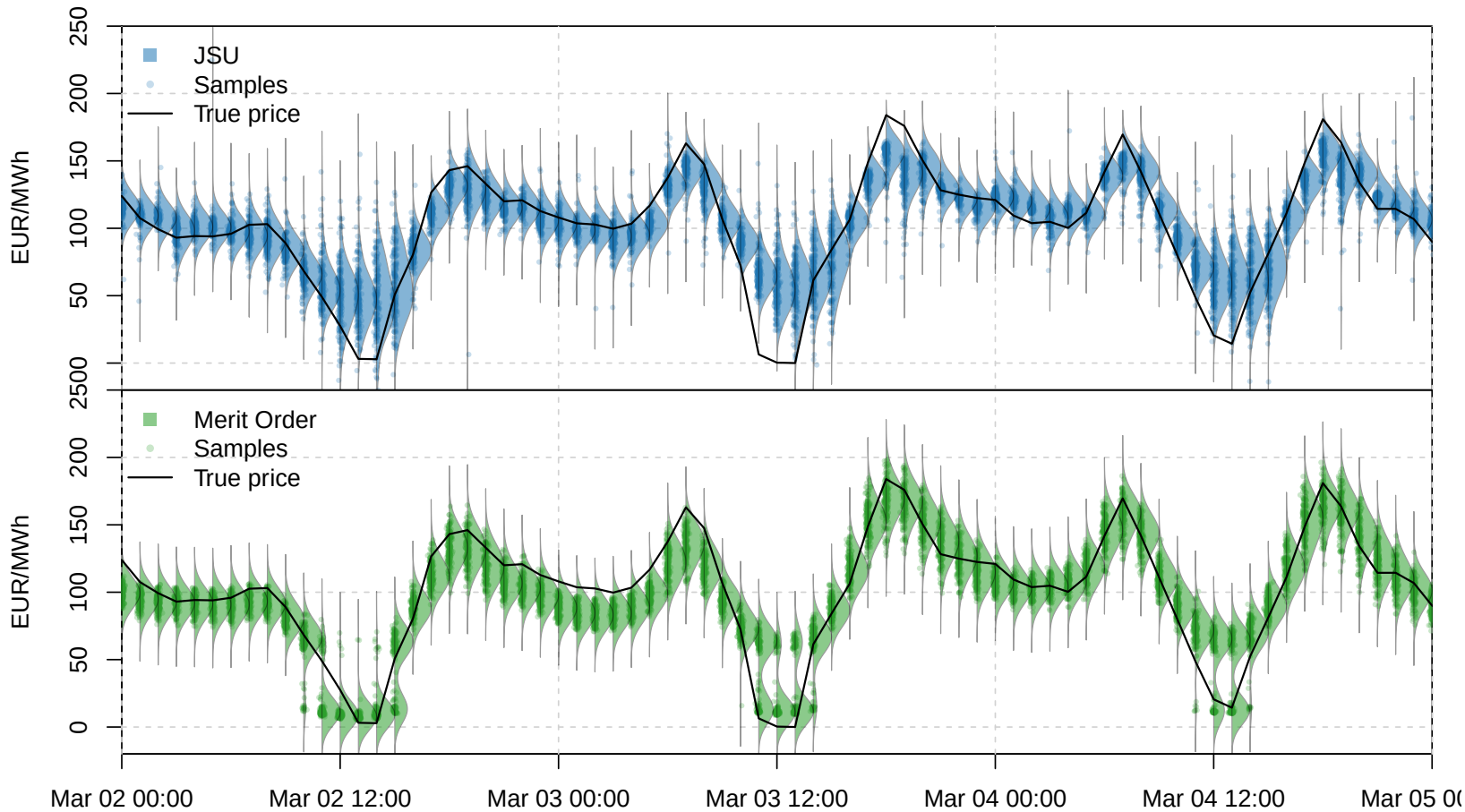
- Efficiencies treated as hyperparameters (weights) to be estimated
- Solved using black-box Bayesian stochastic optimization (R pacakge mlrMBO).

Results

	RMSE	MAE	CRPS
Naive	41.47	25.98	20.46
JSU	29.19	18.21	14.04
Data-Driven MO	25.81	17.28	**13.92**

Table 2: Model Error Comparison

Distribution



Conclusions & Relevance

Data-driven merit order model:

- is a fundamental model at its core $\Rightarrow$ 100% interpretable, transparent, explainable
- captures complex dynamics, e.g: price clusters volatility
- extended to probabilistic:d captures complex non-parametric distributions

Link

- Previous work:

A data-driven merit order:
Learning a fundamental electricity price model
Energy Economics 2025

https://arxiv.org/abs/2501.02963

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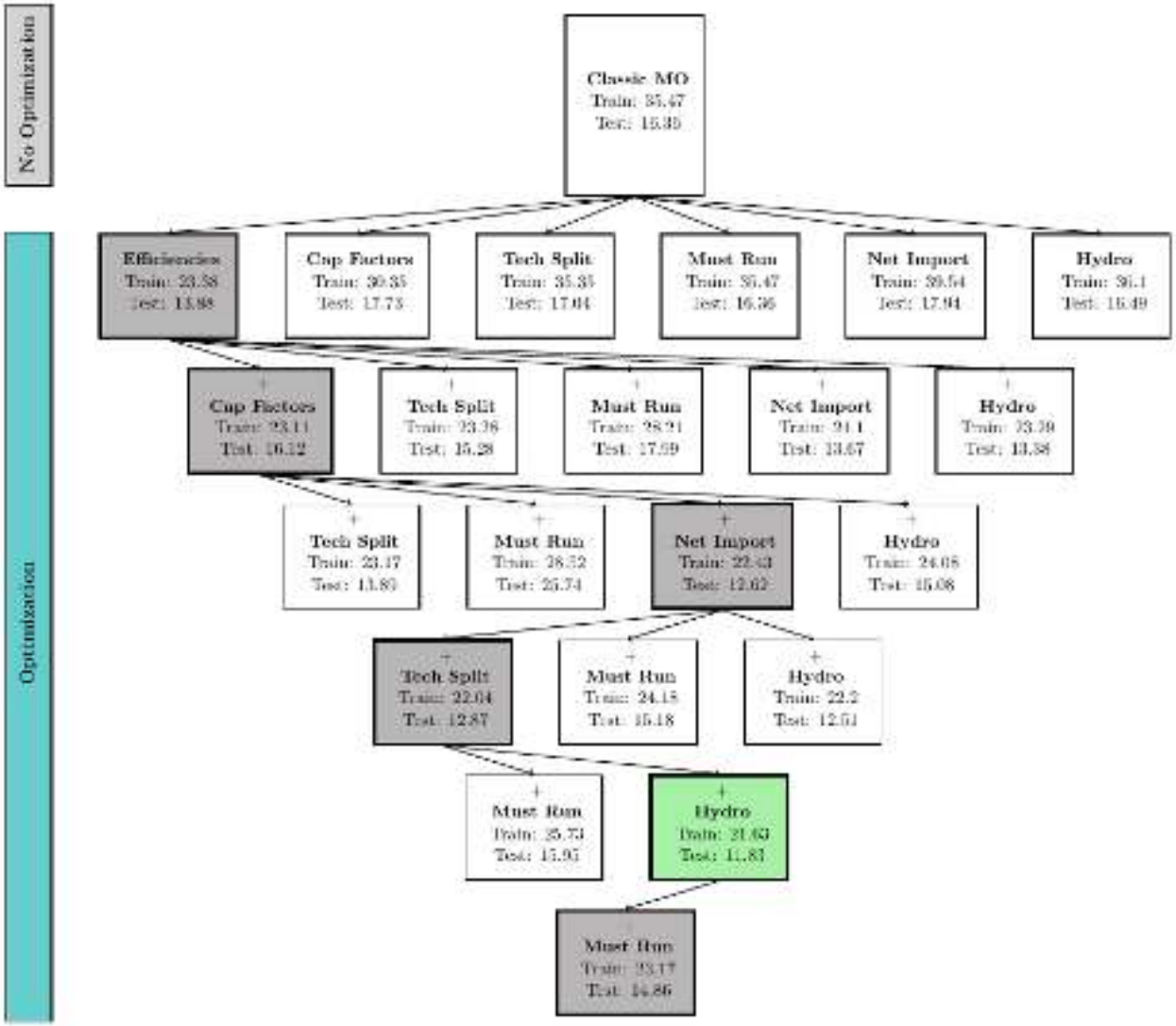
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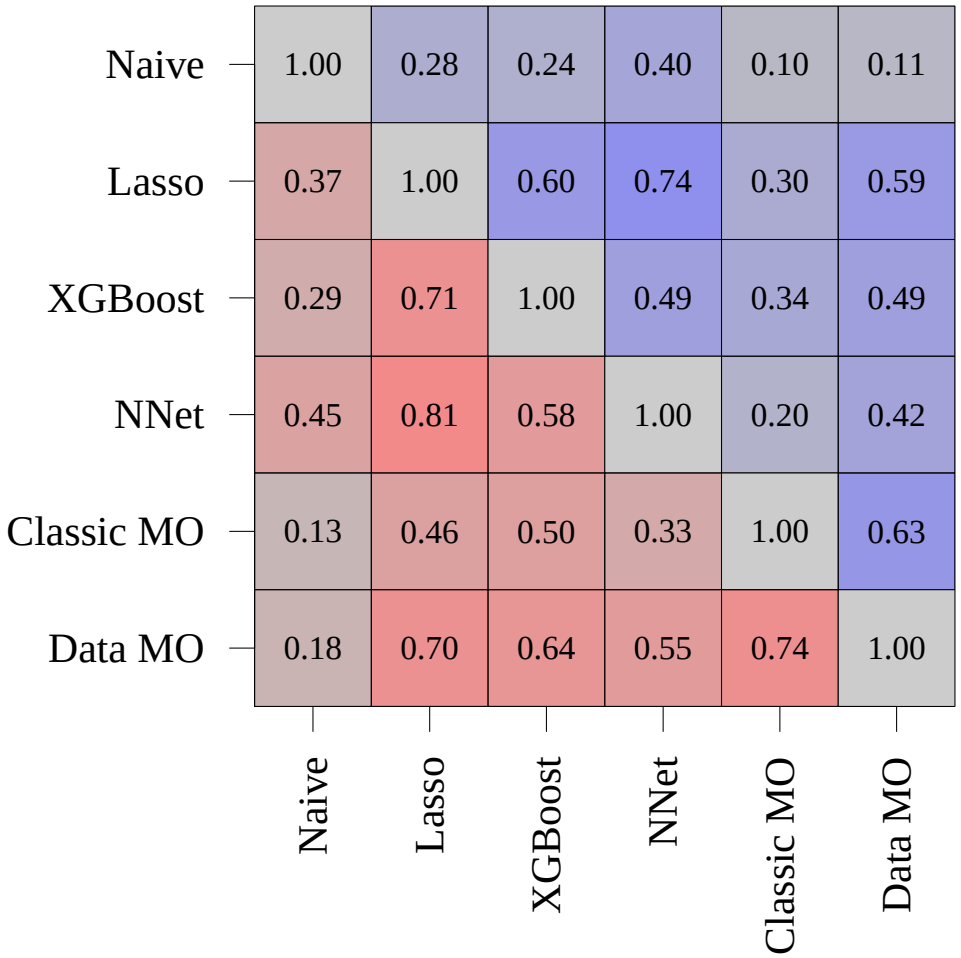
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Backup slides

MO forward selection tree

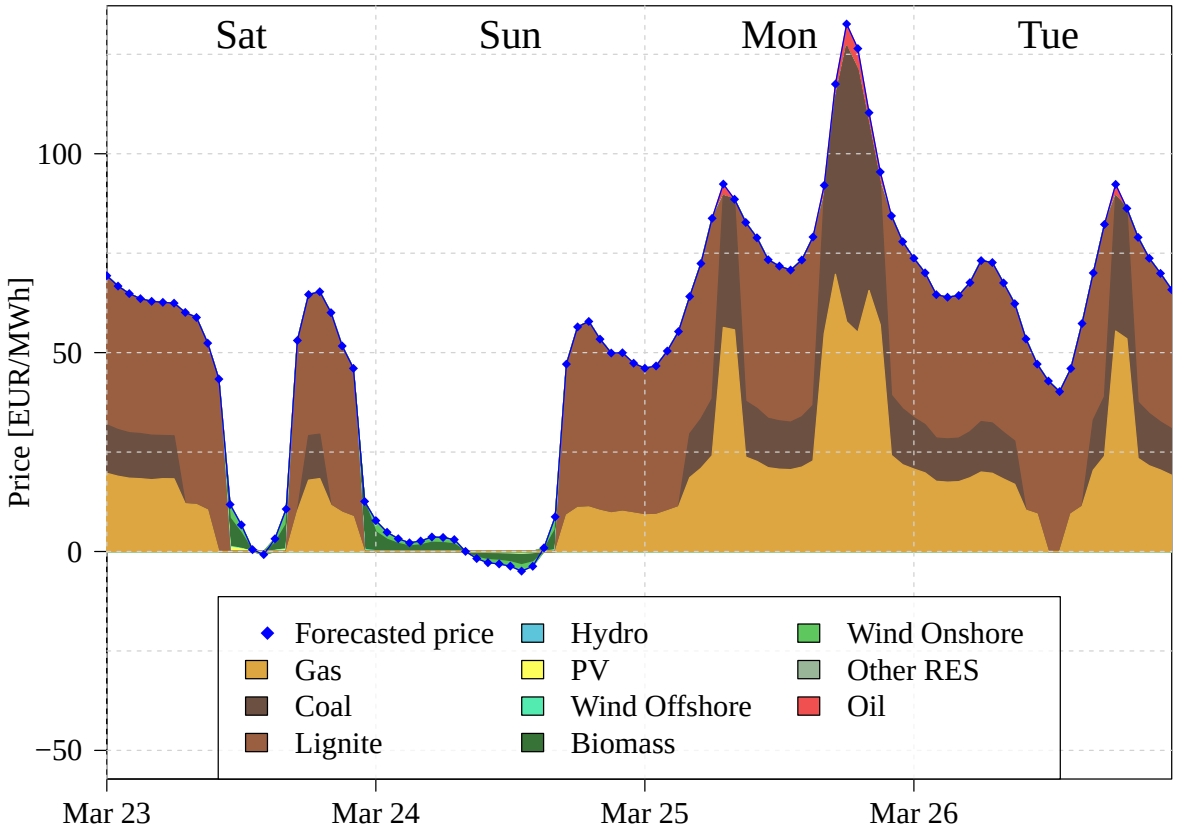


Correlation of errors

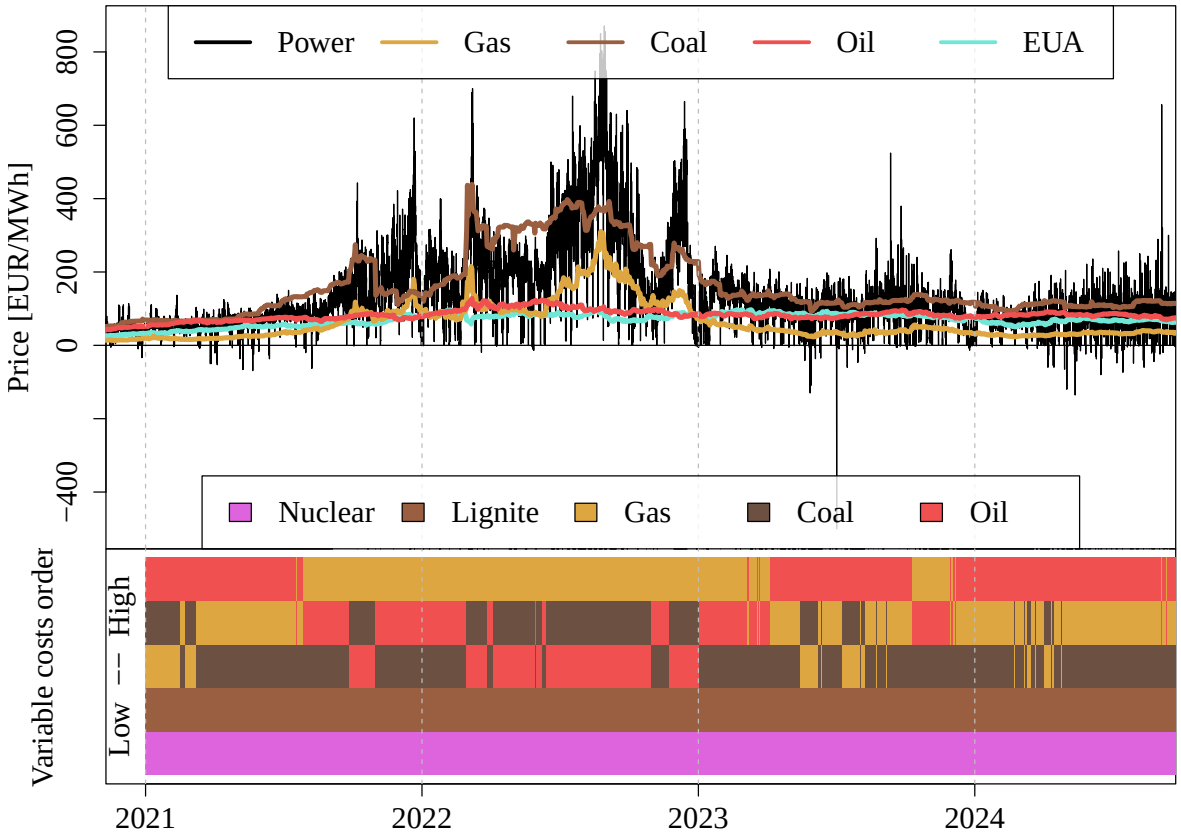


- Pearson's correlation on the lower triangle, distance correlation on the upper triangle.

Marginal costs



Fuel switches



Hydro model

$$\text{Hydro}_{d,h} = \beta_0 + \underbrace{\sum_{s=1}^{11} \beta_{\text{cross},s}\text{Hydro}_{d-1,s} + \sum_{s=12}^{24} \beta_{\text{cross},s}\text{Hydro}_{d-2,s}}_{\text{Cross-hour lags}} + \underbrace{\sum_{l=2}^{14} \beta_{\text{lag},l}\text{Hydro}_{d-l,h}}_{\text{Same hour lags}} \tag{3}$$

$$+ \underbrace{\sum_{i=1}^{4} \beta_{\text{SoY},i}\text{Season}^{(i)}(d,h)}_{\text{Annual seasonality}} + \underbrace{\sum_{i=1}^{7} \beta_{\text{DoW},i}\text{WeekDay}_d^{(i)}}_{\text{Weekly Seasonality}} + \varepsilon_t \quad \forall \quad h \in \{0,\ldots,23\}$$

where d is the day and h the hour. The model is estimated using LASSO and the optimal lambda parameter is chosen by the BIC (Bayesian information criterion).

Net import model

$$\begin{aligned}\text{NetImport}_{d,h}^{\text{bznHome}} = \beta_0 &+ \underbrace{\sum_{\text{bzn}\in\mathcal{B}} \beta_{\text{Load}}^{\text{bzn}} \text{LoadDA}_{d,h}^{\text{bzn}}}_{\text{Day-ahead load}} + \underbrace{\sum_{\text{bzn}\in\mathcal{B}} \sum_{pl\in\mathcal{R}} \beta_{\text{pl, RES}}^{\text{bzn}} \text{RESDA}_{pl,d,h}}_{\text{Renewables day-ahead}} \\ &+ \underbrace{\sum_{\text{bzn}\in\mathcal{B}} \sum_{pl\in\mathcal{C}} \beta_{\text{pl, gen}}^{\text{bzn}} \text{Generation}_{pl,(d-1)\mathbf{1}_{h\in\{1,\dots,11\}}+(d-2)\mathbf{1}_{h\in\{12,\dots,24\}}}}_{\text{Generation of non-RES}} \\ &+ \underbrace{\sum_{\text{i}=1}^{14} \beta_{\text{lag},i} \text{NetImport}_{d-i,(d-1)\mathbf{1}_{h\in\{1,\dots,11\}}+(d-2)\mathbf{1}_{h\in\{12,\dots,24\}}}^{\text{bznHome}}}_{\text{Lagged net imports}} \\ &+ \varepsilon_t \quad \forall \quad h \in \{0,\dots,23\}\end{aligned} \tag{4}$$

where d is the day, is h the hour, $\mathcal{B}$ is the set of all bidding zones of physical flows to and from and including the modelled zone bznHome, $\mathcal{R}$ is the set of all renewable power plants, $\mathcal{C}$ is the set of all conventional power plants, and $\mathbf{1}$ is the indicator function.

Expert model

$$\text{Price}_{d,h} = \beta_0 + \underbrace{\sum_{i=1}^{14} \beta_{\text{lag},i}\text{Price}_{d-i,h}}_{\text{Autoregressive effects}} + \underbrace{\beta_{\max}\max(\text{Price}_{d-1,\cdot}) + \beta_{\min}\min(\text{Price}_{d-1,\cdot})}_{\text{Non-linear effects: min and max of yesterday}} \tag{5}$$

$$+ \underbrace{\beta_{\text{last}}\text{Price}_{d-1,23}}_{\text{Cross-period effects}} + \underbrace{\beta_{\text{load}}\text{LoadDA}_{d,h}}_{\text{Load day-ahead forecast}} + \underbrace{\beta_{\text{res}}\text{RESDA}_{d,h}}_{\text{RES day-ahead forecast}}$$

$$+ \underbrace{\beta_{\text{gas}}\text{Gas}_{d-2} + \beta_{\text{coal}}\text{Coal}_{d-2} + \beta_{\text{oil}}\text{Oil}_{d-2} + \beta_{\text{CO}_2}\text{CO}_{2,d-2}}_{\text{Fuel prices}}$$

$$+ \underbrace{\sum_{i=1}^{7} \beta_{\text{DoW},i}\text{WeekDay}^{(i)}(d)}_{\text{Weekly seasonality}} + \underbrace{\sum_{i=1}^{4} \beta_{\text{SoY,i}}\text{Season}^{(i)}(d,h)}_{\text{Annual seasonality}} + \varepsilon_t \quad \forall \quad h \in \{0,\ldots,23\}$$

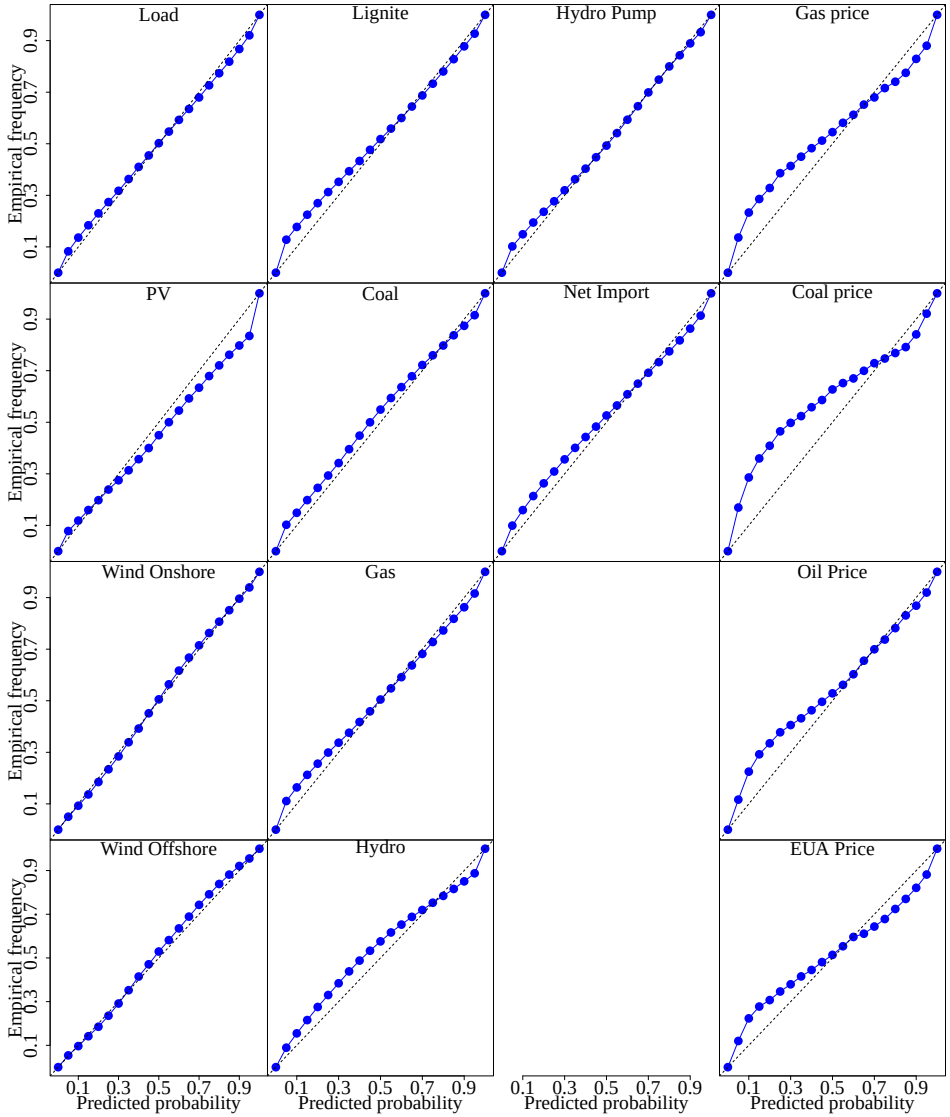
where d is the day and h the hour.

Parameter estimates

Parameter	Lower (Θ^L)	Upper (Θ^U)	Classic MO (Θ_{init})	Data MO + Efficiencies	+ Cap Factor	+ Net Import	+ Tech Split	+ Hydro	+ Must Run
η^L_{gas}	0.10	0.50	0.25	0.40	0.42	0.29	0.33	0.23	0.33
η^L_{coal}	0.10	0.50	0.35	0.12	0.14	0.26	0.13	0.17	0.11
$\eta^L_{lignite}$	0.10	0.50	0.30	0.22	0.25	0.44	0.31	0.33	0.21
η^L_{oil}	0.10	0.50	0.24	0.20	0.15	0.33	0.32	0.45	0.11
$\eta^L_{nuclear}$	0.10	0.50	0.32	0.37	0.25	0.46	0.19	0.47	0.10
η^L_{gas2}	0.10	0.50	0.10	0.44	0.40	0.37	0.32	0.24	0.47
η^U_{gas}	0.10	1.00	0.40	0.71	0.68	0.72	0.54	0.64	0.42
η^U_{coal}	0.10	1.00	0.46	0.20	0.64	0.49	0.51	0.52	0.57
$\eta^U_{lignite}$	0.10	1.00	0.43	0.69	0.63	0.80	0.72	0.68	0.52
η^U_{oil}	0.10	1.00	0.44	0.36	0.16	0.65	0.74	0.78	0.19
$\eta^U_{nuclear}$	0.10	1.00	0.42	0.43	0.58	0.51	0.31	0.83	0.50
η^U_{gas2}	0.10	1.00	0.20	0.78	0.56	0.68	0.49	0.62	0.53
$b^L_{resHydro}$	-500.00	0.00	-500.00	-175.25	-456.42	-422.35	-51.41	-464.70	-28.07
b^L_{resPV}	-500.00	0.00	-500.00	-468.87	-474.47	-224.24	-494.73	-364.57	-488.55
$b^L_{resWindOn}$	-500.00	0.00	-70.00	-1.45	-416.73	-180.61	-159.54	-121.73	-110.47
$b^L_{resWindOff}$	-500.00	0.00	-150.00	-38.83	-203.47	-75.37	-385.07	-440.78	-13.28
$b^L_{resBiomass}$	-500.00	0.00	-200.00	-95.90	-12.56	-201.79	-2.76	-24.87	-262.39
$b^L_{resOther}$	-500.00	0.00	-500.00	-497.95	-142.49	-177.27	-21.66	-103.21	-476.19
$b^U_{resHydro}$	0.00	20.00	20.00	12.67	7.52	19.04	17.89	17.91	17.68
b^U_{resPV}	0.00	20.00	20.00	13.71	9.41	12.93	4.51	17.64	19.48
$b^U_{resWindOn}$	0.00	20.00	20.00	18.14	18.90	14.31	15.07	11.89	9.53
$b^U_{resWindOff}$	0.00	20.00	20.00	11.95	11.79	4.68	0.01	12.91	0.40
$b^U_{resBiomass}$	0.00	20.00	20.00	15.07	9.63	3.36	1.79	13.93	4.68
$b^U_{resOther}$	0.00	20.00	20.00	4.70	2.09	14.98	3.97	2.44	18.12
cf_{gas}	1.00	2.00	1.50	1.50	1.66	1.62	1.63	1.34	1.52
cf_{coal}	1.00	2.00	1.50	1.50	1.23	1.13	1.95	1.28	1.70
$cf_{lignite}$	1.00	2.00	1.50	1.50	1.11	1.03	1.25	1.04	1.35
cf_{oil}	1.00	2.00	1.50	1.50	1.75	1.43	1.84	1.76	1.86
$cf_{resHydro}$	1.00	2.00	0.00	0.00	0.00	0.00	0.00	1.10	1.53
$cf_{netImport}$	0.00	1.00	0.00	0.00	0.00	0.54	0.39	0.55	0.50
$cf_{resBiomass}$	1.00	2.00	1.50	1.50	1.35	1.20	1.16	1.45	1.01
cf_{gas2}	1.00	2.00	1.50	1.50	1.96	1.71	1.04	1.08	1.27
mr_{gas}	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.28
mr_{coal}	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.06
$mr_{lignite}$	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02
mr_{oil}	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.43
$mr_{nuclear}$	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.54
$mr_{resHydro}$	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.73
$mr_{resBiomass}$	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.12
mr_{gas2}	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.10
gs	0.00	1.00	0.00	0.00	0.00	0.00	0.58	0.41	0.50

Table 3: Parameter bounds, initial values, and optimized values for Θ_{ext} across various model setups, following the optimal forward selection path in Figure **??**. Each column corresponds to a different model setup and optimization, with one variable added at each step. The right-most column represents the full model, where all six parameter types are optimized simultaneously.

Calibration of inputs distributions



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